

123456

A

μC

File: μC.kicad_sch

B

Power

File: power.kicad_sch

C

In_Out

File: in_out_puts.kicad_sch

D

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Sheet: /
File: LaserOps_PCB.kicad_sch

Title: Nerf-ATG

Size: A4Date: 2025-04-13

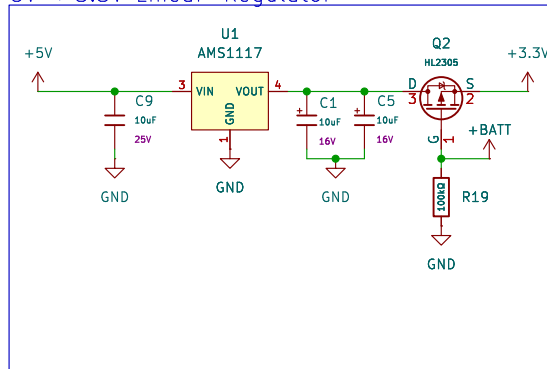
Rev: 1.0

KiCad E.D.A. 9.0.0Id: 1/4

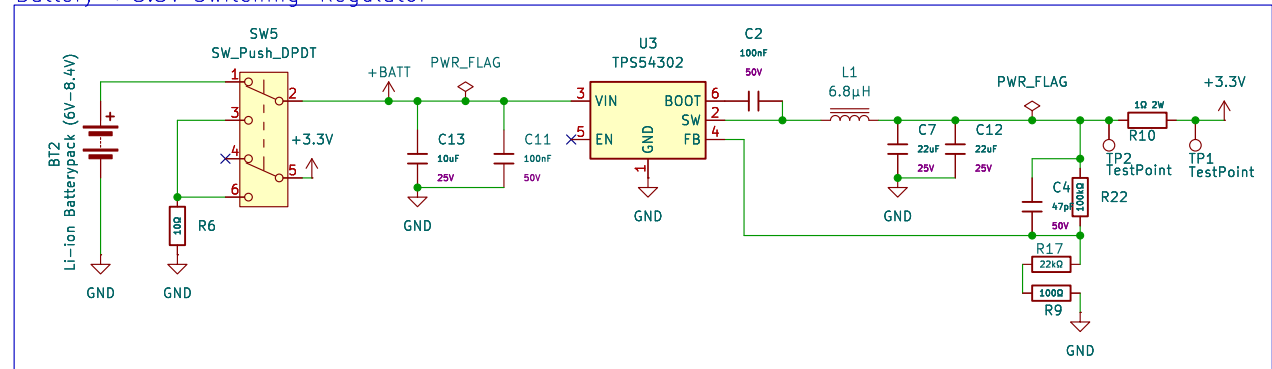
123456

By default, everything is configured for the Nerf Laser Ops Deltaburst Blaster. By cutting off the solder jumpers and re-soldering, the board can also be configured for the Nerf Laser Ops Alphapoint Blaster. In addition, there are some new features that are optional, such as the camera(OV5060), an Accelorometer or an OLED display (SSD1306).

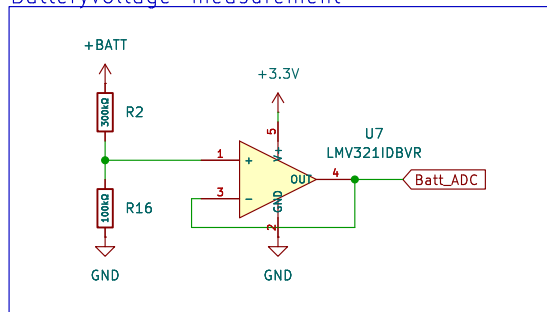
5V->3.3V Linear-Regulator



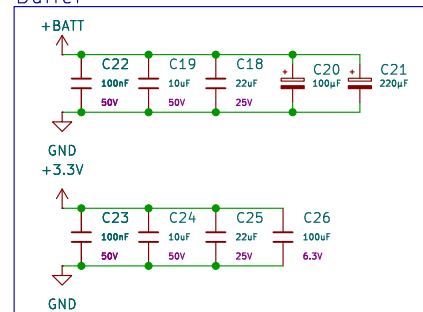
Battery->3.3V Switching-Regulator



Batteryvoltage-measurement



Buffer



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Sheet: /Power/
File: power.kicad_sch

Title: Nerf-ATG

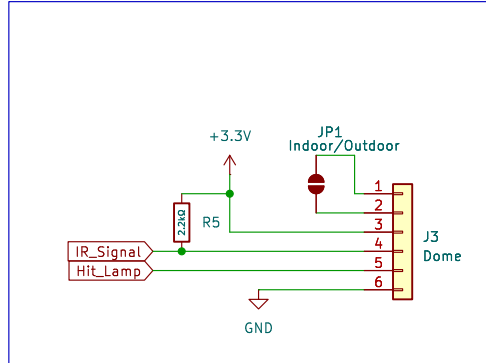
Size: A4 Date: 2025-04-13

KiCad E.D.A. 9.0.0

Rev: 1.0

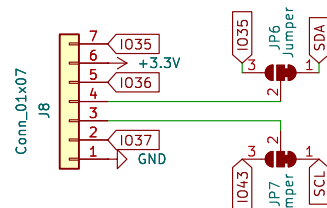
Id: 2/4

IR-Reciver_Connector

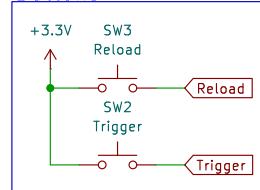


Status-visualization

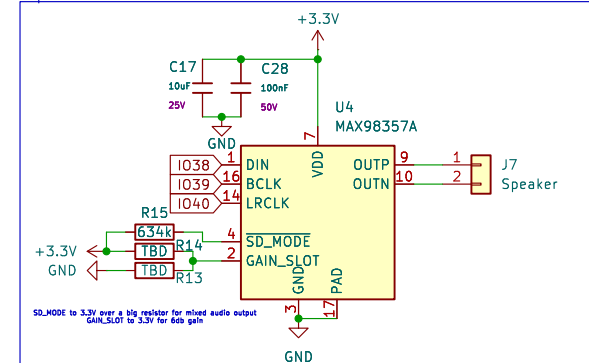
Connectable devices:
Displays:
- Deltaburst display
- SSD1306
LED-boards:
- Alphapoint status-LED



Buttons

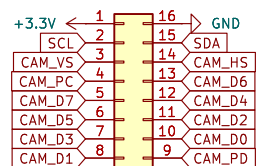


Speaker



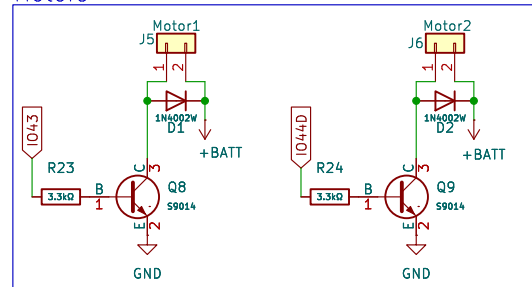
CAM_Interface

Camera_Interface1
Conn_02x08_Counter_Clockwise

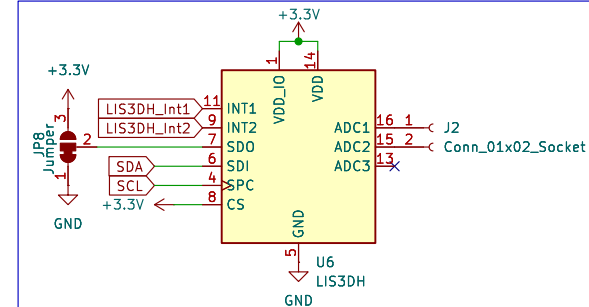


Interface for Adafruit
OV5640 Camera Breakouts

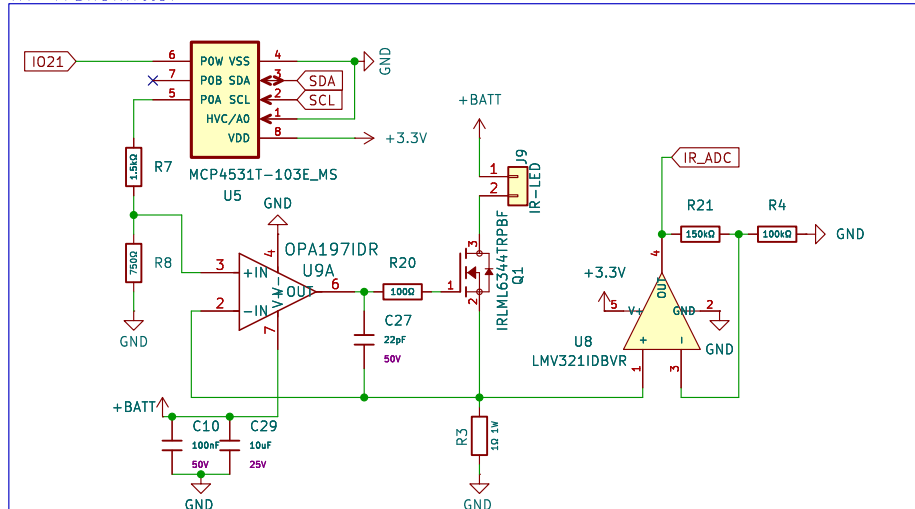
Motors



Accelerometer



IR-Transmitter



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Sheet: /In_Out/

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Size: A4

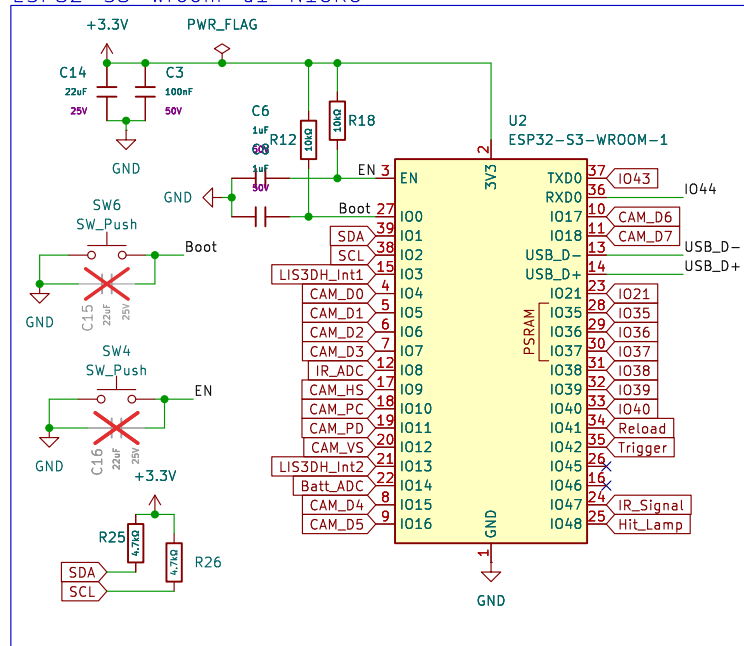
Date: 2025-04-13

KiCad E.D.A. 9.0.0

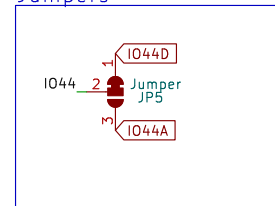
Rev: 1.0

Id: 3/4

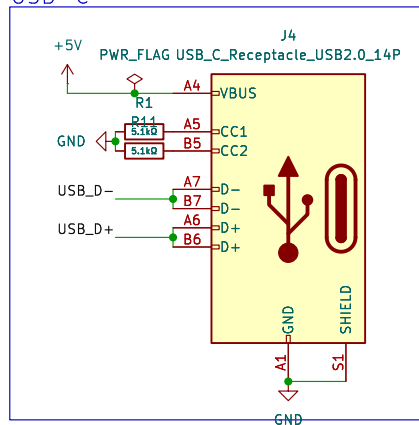
ESP32-S3-wroom-u1-N16R8



Jumpers



USB-C



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Rev: 1.0
Id: 4/4